

# HS4002: Lecture 5B

## Linear Models and Ordinary Least Squares Regression

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# Hypothesis Testing

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# What is a hypothesis test?

Any statistical hypothesis test, no matter how complex, is based on the logic of stochastic proof by contradiction.

# What is a hypothesis test?

- In mathematics, proof by contradiction is a proof technique where we begin by assuming the validity of a hypothesis we would like to disprove, and then derive a contradiction under the same hypothesis.

# What is a hypothesis test?

- Hypothesis testing uses the same logic of proof by contradiction and yet there is uncertainty, hence the word “stochastic.”
  - Specifically, we can never conclude, with certainty, that a hypothesis is incorrect.
  - Instead, we argue that the hypothesis is likely to be incorrect.

# What is a P-Value?

A p-value is the probability under a specified statistical model that a statistical summary of the data would be equal to or more extreme than its observed value.

- P-values indicate how incompatible the data are with a specified statistical model (e.g. the null hypothesis).
  - e.g. they tell us the likelihood that we collect the data as extreme as we do, if the null were true.
- A p-value does not measure the size of an effect or the importance of a result.



# One-tailed vs. two-tailed tests

- “More extreme” depends on how we specify the alternative hypothesis.
- A two-tailed (two-sided) test asks whether the parameter differs from the null value in either direction.
  - e.g.  $H_0 : r = 0$  vs.  $H_A : r \neq 0$
  - Extreme outcomes sit in both the left and right tails of the null distribution.
- A one-tailed (one-sided) test asks whether the parameter differs in one pre-specified direction.
  - e.g.  $H_0 : r = 0$  vs.  $H_A : r > 0$  (right-tailed), or  $H_A : r < 0$  (left-tailed)

# One-tailed vs. two-tailed tests

- For the same observed statistic, a one-tailed p-value is typically half the two-tailed p-value—but only if the estimate falls in the direction you specified in advance.
- If the estimate goes the other way, the one-tailed test does not count that as evidence against  $H_0$ .
- In practice, most software defaults to two-tailed tests (e.g. `scipy.stats.pearsonr`, OLS  $t$ -tests with  $P > |t|$ ).
- Use a one-tailed test only when theory and your research question genuinely rule out one direction before seeing the data.

## An Example

Imagine that we are testing the statistical significance of the Pearson's  $r$  between income and occupational prestige.

# An example

Pearson's  $r$  between income and occupational prestige.

```
from scipy import stats

r, p = stats.pearsonr(df['income_cont'], df['prestg10'])
print(f"Pearson r = {r:.4f}, p-value = {p:.3g}, n = {len(df)}")

ci = stats.pearsonr(df['income_cont'], df['prestg10']).confidence_interval()
print(f"95% CI = [{ci.low:.4f}, {ci.high:.4f}]")

Pearson r = 0.4185, p-value = 2.42e-31, n = 707
95% CI = [0.3557, 0.4775]
```

## An Example

We estimate an  $\hat{r}$  of 0.42, with a p-value smaller than 0.001.

1. The null hypothesis is that the true value of  $r = 0$ ; the alternative is two-tailed ( $r \neq 0$ ).
2. The p-value tells us that there is a probability of  $< 0.001$  that we obtain an estimated  $|\hat{r}|$  of 0.42 or more extreme if the true value of  $r = 0$ .
3. This seems quite unlikely under the null, so we reject  $H_0$ .

Cramer's  $V$  between race and highest degree attained.

```
contingency = pd.crosstab(df['race'], df['degree'])

res = stats.chi2_contingency(contingency)
v = stats.contingency.association(contingency, method='cramer')

print(f"chi2 = {res.statistic:.4f}, dof = {res.dof}, p-value = {res.pvalue:.4f}")
print(f"Cramer's V = {v:.4f}, n = {len(df)}")

chi2 = 26.0452, dof = 8, p-value = 0.0010
Cramer's V = 0.1357, n = 707
```

# Class Exercise B

## Linear regression.

```
import statsmodels.formula.api as smf

model = smf.ols('income_cont ~ age + working + sex_woman + prestg10', data=df).fit()
print(model.summary())
```

### OLS Regression Results

```
=====
Dep. Variable:          income_cont  R-squared:                0.208
Model:                  OLS          Adj. R-squared:            0.204
Method:                 Least Squares  F-statistic:              46.22
Date:                  Tue, 09 Sep 2026  Prob (F-statistic):        1.73e-34
No. Observations:      707          Log-Likelihood:            -8827.9
Df Residuals:          702          AIC:                    1.767e+04
Df Model:              4

=====
```

|                    | coef       | std err  | t      | P> t  | [0.025    | 0.975]    |
|--------------------|------------|----------|--------|-------|-----------|-----------|
| Intercept          | -4.096e+04 | 1.34e+04 | -3.051 | 0.002 | -6.73e+04 | -1.46e+04 |
| working[T.Working] | 2.825e+04  | 5763.704 | 4.901  | 0.000 | 1.69e+04  | 3.96e+04  |
| sex_woman[T.Woman] | -1.058e+04 | 4849.897 | -2.181 | 0.030 | -2.01e+04 | -1055.371 |
| age                | 295.7130   | 165.027  | 1.792  | 0.074 | -28.292   | 619.718   |
| prestg10           | 2134.3988  | 181.307  | 11.772 | 0.000 | 1778.429  | 2490.368  |

```
=====
```

**What is a Linear Model? What is an OLS estimator?**

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# Intuitions Behind the Linear Model

Intuitively:

$$\text{outcome}_i = \text{model} + \text{error}_i$$

We usually use  $Y_i$  to represent the outcome variable for unit  $i$ , the model comprises a set of explanatory variables, each weighted by a  $\beta$  regression coefficient, and an error term containing the unexplained variance in the outcome.

Simple linear models:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \epsilon_i$$

where  $Y$  is the dependent variable,  $X_1 \dots X_k$  are the set of independent variables included in the model,  $\beta_0 \dots \beta_k$  are the beta coefficients that weigh each independent variable, and  $\epsilon_i$  is the error term.

# What is the OLS estimator?

- The Ordinary Least Square (OLS) estimator is one way of obtaining estimates of  $\beta_0, \dots, \beta_k$ .
  - $\beta_0, \dots, \beta_k$  are unknown population parameters.
  - Following convention, we will be adding a little hat to every estimate (e.g.  $\hat{\beta}_k$  is an estimate of  $\beta_k$ )

# What is the OLS estimator?

- To produce estimates of  $\hat{\beta}_0 \dots \hat{\beta}_k$ , we regress  $Y$  on  $X_1 \dots X_k$ .
- The OLS estimator chooses values of  $\hat{\beta}_0 \dots \hat{\beta}_k$  that best minimizes the sum of squared difference between the outcome variables predicted by the model, and the actual outcomes observed.

## Interpreting $\beta$ Coefficients

$\beta$  coefficients are partial effects. That is to say, they have ceteris paribus interpretations.

Suppose I regress income on age and a bunch of other covariates, and estimate a  $\hat{\beta}_{\text{age}}$  of 500. This would tell me that one unit increase in age (1 year) would result in an increase in income of  $500 \times \$1$ , holding all the other variables constant.

- $\hat{\beta}_j$  is an estimate with a known distribution.
- This means we know their variance. We can calculate the p-value of an estimate, assuming a given statistical model.

$$\hat{\beta}_j \sim N(\beta_j, V(\hat{\beta}_j))$$

$$\frac{\hat{\beta}_j - \beta_j}{s_{\hat{\beta}_j}} \sim N(0, 1)$$

# Standard Hypothesis Testing of $\hat{\beta}_j$

We test the following null hypothesis:

$$H_0 : \beta_j = 0 \quad \text{vs.} \quad H_A : \beta_j \neq 0$$

We conduct a two-tailed test: either a sufficiently large positive or negative  $\hat{\beta}_j$  counts as evidence against the null. That is what the usual  $P > |t|$  column reports.



## Comparing Goodness-of-fit

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The  $R^2$  of the regression, sometimes called the coefficient of determination, is defined as

$$R^2 = SSE/SST = 1 - (SSR/SST).$$

This is interpreted as the proportion of the sample variation in  $Y$  that is explained by the regression model. The higher it is, the "better" the model.

- Total sum of squares (SST)
  - $SST = \sum (Y_i - \bar{Y})^2$
  - $SST = SSE + SSR$ .
- Explained sum of squares (SSE)
  - $SSE = \sum (\hat{Y}_i - \bar{Y})^2$
- Residual sum of squares (SSR)
  - $SSR = \sum \epsilon_i^2 = \sum (Y_i - \hat{Y})^2$

Similar to  $R^2$ , plus a penalty for adding additional independent variables:

$$\bar{R}^2 = 1 - \frac{SSR/(n - k - 1)}{SST/(n - 1)},$$

where  $k$  is the no. of variable, and  $n$  is the no. of observations. This corrects one problem with  $R^2$  values, which is that they increase monotonically with the number of independent variables added (it's always better to add more). The higher it is, the "better" the model.

## Akaike Information Criterion (AIC)

The Akaike information criterion (AIC) is another measure of fit which penalizes the model for having more variables:

$$AIC = n \ln\left(\frac{SSE}{n}\right) + 2k,$$

where  $k$  is the no. of variable, and  $n$  is the no. of observations. AIC only makes sense when you compare models against one another. The larger the AIC value is, the worse the model; the smaller, the better.

We usually use the F-Test to test two models, a smaller restricted model  $r$  and a larger unrestricted model  $ur$  (same variables as restricted +  $q$  more variables),

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k - 1)},$$

where  $k$  is the no. of variables in the restricted case,  $q$  is the no. of additional variables in the unrestricted case, and  $n$  is the no. of observations.

F-statistics have known distributions.

The null hypothesis tests if the  $q$  additional variables in the unrestricted model all have coefficients of zero.

Example: Peterson & Kern's  
(1996) work on omnivorous taste.

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How have the cultural tastes of Americans, in particular Americans with "highbrow tastes," changed from 1982 to 1992?

- Defining highbrows:
  - Someone who likes both classical music and opera, and chooses one of these as best-liked among all kinds of music.
- Key Dependent Variables:
  - No. of low-brow music genres liked.
  - No. of middle-brow music genres liked.

**Figure 1: OLS Results**

| Variables                       | Highbrows                      |       |                                   |      | Others                         |        |                                   |        |
|---------------------------------|--------------------------------|-------|-----------------------------------|------|--------------------------------|--------|-----------------------------------|--------|
|                                 | (Model 1)                      |       | (Model 2)                         |      | (Model 3)                      |        | (Model 4)                         |        |
|                                 | Number of Lowbrow Genres Liked |       | Number of Middlebrow Genres Liked |      | Number of Lowbrow Genres Liked |        | Number of Middlebrow Genres Liked |        |
|                                 | b                              | Beta  | b                                 | Beta | b                              | Beta   | b                                 | Beta   |
| Birth year                      | .02                            | .16** | -.01                              | -.07 | .01                            | .12**  | -.01                              | -.23** |
| Year of interview<br>(1 = 1992) | .44                            | .15*  | .25                               | .13* | .20                            | .07**  | .15                               | .07**  |
| <i>Control Variables</i>        |                                |       |                                   |      |                                |        |                                   |        |
| Male                            | -.07                           | -.02  | -.20                              | -.10 | .01                            | .01    | -.20                              | -.09** |
| Adjusted family income          | .00                            | -.11  | .01                               | .04  | .00                            | .01    | .00                               | .12**  |
| White                           | -.92                           | -.13  | .60                               | .13* | .18                            | .05**  | .38                               | .12**  |
| Education in years              | .05                            | .10   | .00                               | .01  | .05                            | .11**  | .10                               | .28**  |
| Size of community               | -.01                           | -.04  | .00                               | .02  | -.02                           | -.08** | .02                               | .08**  |
| Constant                        | -27.45*                        | —     | 9.25                              | —    | -16.61**                       | —      | 26.20**                           | —      |
| Significance of F               | —                              | .00   | —                                 | .12  | —                              | .00    | —                                 | .00    |
| Adjusted R <sup>2</sup>         | —                              | .06   | —                                 | .02  | —                              | .06    | —                                 | .16    |
| Number of respondents           | —                              | 354   | —                                 | 354  | —                              | 10,967 | —                                 | 10,967 |

\*p < .05    \*\*p < .01 (one-tailed tests)  
 \*p < .05    \*\*p < .01 (two-tailed tests)

Omnivorousness is replacing snobbishness among Americans of highbrow status.

**Thanks for listening :)**